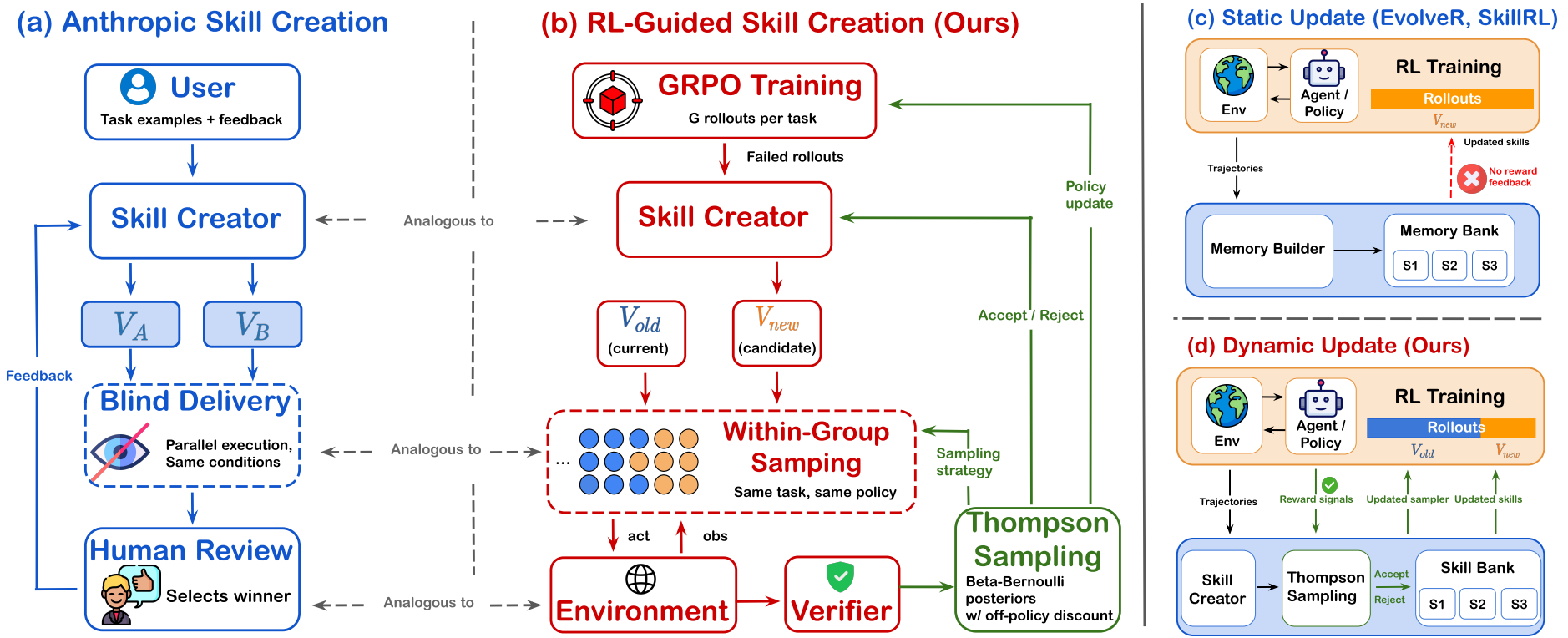
(a) Anthropic Skill Creation
User
Task examples + feedback
Skill Creator
V_A
V_B
Feedback
Blind Delivery
Parallel execution, Same conditions
Human Review
Selects winner
Analogous to
Analogous to
Analogous to
(b) RL-Guided Skill Creation (Ours)
GRPO Training
G rollouts per task
Failed rollouts
Skill Creator
V_{old} (current)
V_{new} (candidate)
Within-Group Samping
Same task, same policy
act
obs
Environment
Verifier
Thompson Sampling
Beta-Bernoulli posteriors w/ off-policy discount
Sampling strategy
Accept / Reject
Policy update
(c) Static Update (EvolveR, SkillRL)
Env
Agent / Policy
RL Training
Rollouts
V_{new}
Updated skills
No reward feedback
Trajectories
Memory Builder
Memory Bank
S1
S2
S3
(d) Dynamic Update (Ours)
Env
Agent / Policy
RL Training
Rollouts
V_{old}
V_{new}
Trajectories
Reward signals
Updated sampler
Updated skills
Skill Creator
Thompson Sampling
Accept
Reject
Skill Bank
S1
S2
S3